

# Class 09 Mini Project

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## Background

In this mini-project, we will explore FiveThirtyEight's Halloween Candy data set.

We will use lots of **ggplots**, some basic stats, correlation analysis and PCA to make sense of the landscape of US candy. The task for this **Candy mini project** is to explore a candy dataset to find out answers to these types of questions like 'So what is the top ranked snack-sized Halloween candy? What made some candies more desirable than others? Was it price? Maybe it was just sugar content? Were they chocolate? Did they contain peanuts or almonds?' but most of all the job is to have fun, learn by doing hands on data analysis, and hopefully make this type of analysis less frightening for the future! Let's get started.

## 2. Importing Candy Data

```
candy_file <- "candy-data.csv"
candy = read.csv("candy-data.csv", row.names=1)
head(candy)
```

|              | chocolate | fruity | caramel | peanutyalmondy | nougat | crispedricewafer |
|--------------|-----------|--------|---------|----------------|--------|------------------|
| 100 Grand    | 1         | 0      | 1       | 0              | 0      | 1                |
| 3 Musketeers | 1         | 0      | 0       | 0              | 1      | 0                |
| One dime     | 0         | 0      | 0       | 0              | 0      | 0                |
| One quarter  | 0         | 0      | 0       | 0              | 0      | 0                |
| Air Heads    | 0         | 1      | 0       | 0              | 0      | 0                |
| Almond Joy   | 1         | 0      | 0       | 1              | 0      | 0                |

|              | hard | bar | pluribus | sugarpercent | pricepercent | winpercent |
|--------------|------|-----|----------|--------------|--------------|------------|
| 100 Grand    | 0    | 1   | 0        | 0.732        | 0.860        | 66.97173   |
| 3 Musketeers | 0    | 1   | 0        | 0.604        | 0.511        | 67.60294   |
| One dime     | 0    | 0   | 0        | 0.011        | 0.116        | 32.26109   |
| One quarter  | 0    | 0   | 0        | 0.011        | 0.511        | 46.11650   |
| Air Heads    | 0    | 0   | 0        | 0.906        | 0.511        | 52.34146   |
| Almond Joy   | 0    | 1   | 0        | 0.465        | 0.767        | 50.34755   |

### 2.1 What is in this data set?

According to 538 the columns in the dataset include:

- chocolate: Does it contain chocolate?
- fruity: Is it fruit flavored?
- caramel: Is there caramel in the candy?
- peanutyalmondy: Does it contain peanuts, peanut butter or almonds?
- nougat: Does it contain nougat?
- crispedricewafer: Does it contain crisped rice, wafers, or a cookie component?
- hard: Is it a hard candy?
- bar: Is it a candy bar?
- pluribus: Is it one of many candies in a bag or box?
- sugarpercent: The percentile of sugar it falls under within the data set.
- pricepercent: The unit price percentile compared to the rest of the set.
- winpercent: The overall win percentage according to 269,000 matchups (more on this in a moment).

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

- There are 85 different candy types in this data set.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

- There are 38 fruity candy types in the data set.

## 2.2 What is your favorite candy?

We can find the winpercent value for Twix by using its name to access the corresponding row of the dataset. This is because the dataset has each candy name as rownames (recall that we set this when we imported the original CSV file). For example the code for Twix is:

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

An alternate way to do this is with the help of the dplyr package that we have seen previously:

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Snickers", ]$winpercent
```

```
[1] 76.67378
```

- The winpercent for Snickers is 76.67378.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

- The winpercent value for Kitkat is 76.7686.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

- The winpercent is 49.6535.

### Side Note - the `skimr::skim()` function

There is a useful `skim()` function in the `skimr` package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

- `Skimr()` needs to be installed first.

```
# install.packages("skimr")
library("skimr")
skim(candy)
```

Table 1: Data summary

|                        |       |
|------------------------|-------|
| Name                   | candy |
| Number of rows         | 85    |
| Number of columns      | 12    |
| Column type frequency: |       |
| numeric                | 12    |
| Group variables        | None  |

#### Variable type: numeric

| skim_variable    | n_missing | complete_rate | mean  | sd    | p0    | p25   | p50   | p75   | p100  | hist |
|------------------|-----------|---------------|-------|-------|-------|-------|-------|-------|-------|------|
| chocolate        | 0         | 1             | 0.44  | 0.50  | 0.00  | 0.00  | 0.00  | 1.00  | 1.00  |      |
| fruity           | 0         | 1             | 0.45  | 0.50  | 0.00  | 0.00  | 0.00  | 1.00  | 1.00  |      |
| caramel          | 0         | 1             | 0.16  | 0.37  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| peanutyalmondy   | 0         | 1             | 0.16  | 0.37  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| nougat           | 0         | 1             | 0.08  | 0.28  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| crispedricewafer | 0         | 1             | 0.08  | 0.28  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| hard             | 0         | 1             | 0.18  | 0.38  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| bar              | 0         | 1             | 0.25  | 0.43  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| pluribus         | 0         | 1             | 0.52  | 0.50  | 0.00  | 0.00  | 1.00  | 1.00  | 1.00  |      |
| sugarpercent     | 0         | 1             | 0.48  | 0.28  | 0.01  | 0.22  | 0.47  | 0.73  | 0.99  |      |
| pricepercent     | 0         | 1             | 0.47  | 0.29  | 0.01  | 0.26  | 0.47  | 0.65  | 0.98  |      |
| winpercent       | 0         | 1             | 50.32 | 14.71 | 22.45 | 39.14 | 47.83 | 59.86 | 84.18 |      |

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

- Yes, the **winpercent** column looks to be on a different scale to the majority of the other columns in the data set.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

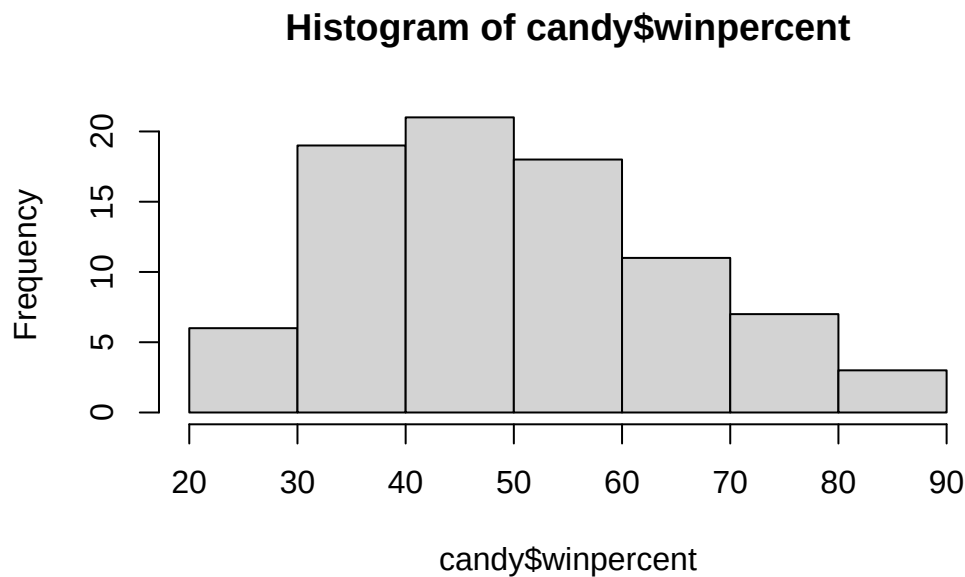
- The zero and one represent if the candy contains chocolate or not. 0 is for if the candy does not contain chocolate and 1 is for if the candy does contain chocolate.

### 3. Exploratory Analysis

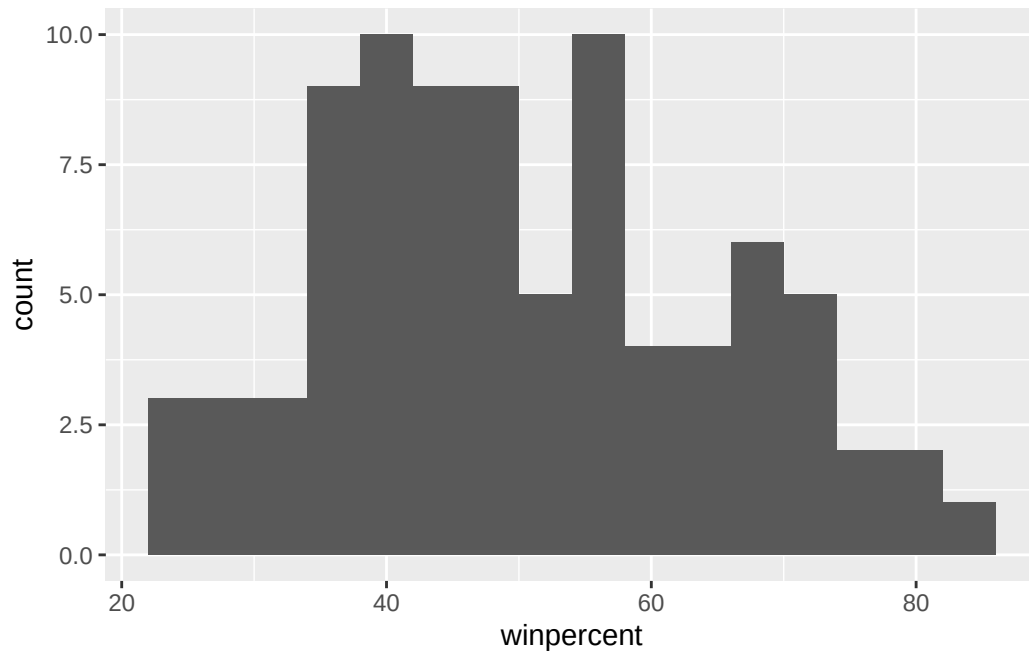
A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`. Either works well in this case and (as always) it's your choice.

Q8. Plot a histogram of `winpercent` values using both base R and ggplot functions.

```
hist(candy$winpercent)
```



```
library(ggplot2)
ggplot(candy, aes(winpercent)) +
  geom_histogram(binwidth = 4)
```



Q9. Is the distribution of winpercent values symmetrical?

- No, from the histograms we can see that the distribution of winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 22.45 | 39.14   | 47.83  | 50.32 | 59.86   | 84.18 |

- The mean of the distribution was above 50% but the median was actually below 50.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[as.logical(candy$chocolate)]) >
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] TRUE
```

- Yes, on average chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
chocolate <- candy$winpercent[as.logical(candy$chocolate)]
fruit <- candy$winpercent[as.logical(candy$fruity)]
t.test(chocolate ,fruit)
```

Welch Two Sample t-test

```
data: chocolate and fruit
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

- Yes, based on the t-test values the difference is statistically significant.

#### 4. Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole dataset by winpercent. Or if you have been getting into the tidyverse and the `dplyr` package you can use the `arrange()` function together with `head()` to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

```
candy |> arrange(winpercent) |> head(5)
```

|                    | chocolate | fruity | caramel | peanutyalmondy | nougat |
|--------------------|-----------|--------|---------|----------------|--------|
| Nik L Nip          | 0         | 1      | 0       | 0              | 0      |
| Boston Baked Beans | 0         | 0      | 0       | 1              | 0      |
| Chiclets           | 0         | 1      | 0       | 0              | 0      |
| Super Bubble       | 0         | 1      | 0       | 0              | 0      |
| Jawbusters         | 0         | 1      | 0       | 0              | 0      |

|  | crispedrice | wafer | hard bar | pluribus | sugarpercent | pricepercent |
|--|-------------|-------|----------|----------|--------------|--------------|
|--|-------------|-------|----------|----------|--------------|--------------|



|                    |   |   |   |   |       |       |
|--------------------|---|---|---|---|-------|-------|
| Nik L Nip          | 0 | 0 | 0 | 1 | 0.197 | 0.976 |
| Boston Baked Beans | 0 | 0 | 0 | 1 | 0.313 | 0.511 |
| Chiclets           | 0 | 0 | 0 | 1 | 0.046 | 0.325 |
| Super Bubble       | 0 | 0 | 0 | 0 | 0.162 | 0.116 |
| Jawbusters         | 0 | 1 | 0 | 1 | 0.093 | 0.511 |

|                    | winpercent |
|--------------------|------------|
| Nik L Nip          | 22.44534   |
| Boston Baked Beans | 23.41782   |
| Chiclets           | 24.52499   |
| Super Bubble       | 27.30386   |
| Jawbusters         | 28.12744   |

- The 5 least liked are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |> arrange(desc(winpercent)) |> head(5)
```

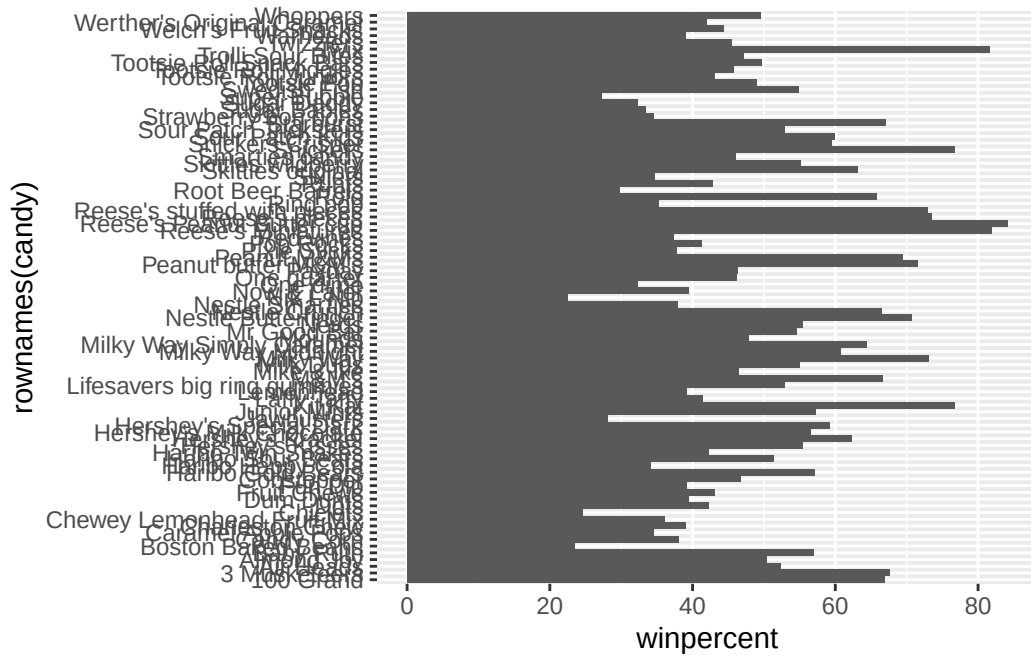
|                           | chocolate | fruity   | caramel    | peanut   | almond   | nougat |
|---------------------------|-----------|----------|------------|----------|----------|--------|
| Reese's Peanut Butter cup | 1         | 0        | 0          |          | 1        | 0      |
| Reese's Miniatures        | 1         | 0        | 0          |          | 1        | 0      |
| Twix                      | 1         | 0        | 1          |          | 0        | 0      |
| Kit Kat                   | 1         | 0        | 0          |          | 0        | 0      |
| Snickers                  | 1         | 0        | 1          |          | 1        | 1      |
|                           | crisped   | rice     | wafer      | hard bar | pluribus | sugar  |
| Reese's Peanut Butter cup |           | 0        | 0          | 0        | 0        | 0.720  |
| Reese's Miniatures        |           | 0        | 0          | 0        | 0        | 0.034  |
| Twix                      |           | 1        | 0          | 1        | 0        | 0.546  |
| Kit Kat                   |           | 1        | 0          | 1        | 0        | 0.313  |
| Snickers                  |           | 0        | 0          | 1        | 0        | 0.546  |
|                           | price     | percent  | winpercent |          |          |        |
| Reese's Peanut Butter cup | 0.651     | 84.18029 |            |          |          |        |
| Reese's Miniatures        | 0.279     | 81.86626 |            |          |          |        |
| Twix                      | 0.906     | 81.64291 |            |          |          |        |
| Kit Kat                   | 0.511     | 76.76860 |            |          |          |        |
| Snickers                  | 0.651     | 76.67378 |            |          |          |        |

- The top 5 all time favorite candy types are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat and Snickers.

Now, let's make a bar plot to visualize the overall rankings in a step-wise fashion.

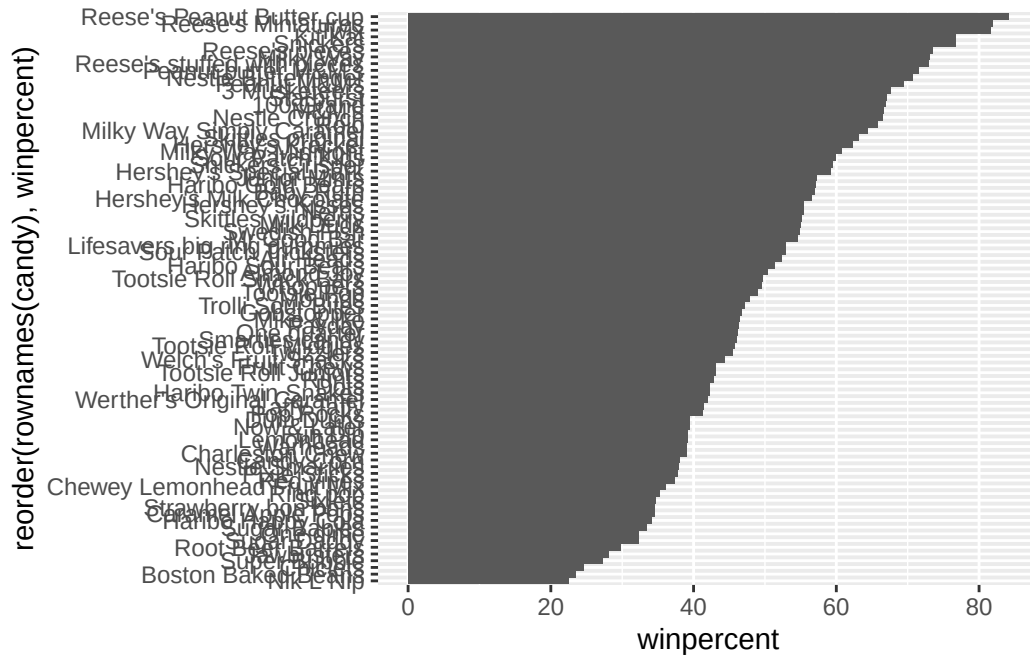
Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)
ggplot(candy) + aes(winpercent, rownames(candy)) + geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
ggplot(candy) + aes(winpercent, reorder(rownames(candy), winpercent)) + geom_col()
```



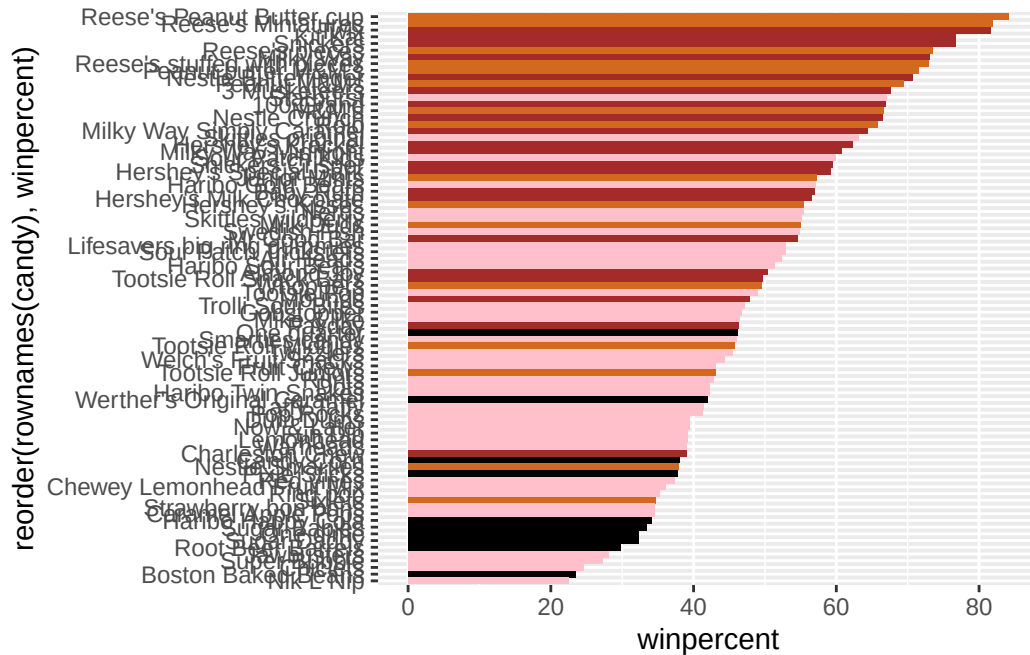
### Time to Add some useful color

We start by setting up a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

Now let's try our barplot with these colors:

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

- The worst ranked chocolate candy is **Sixlets**.

Q18. What is the best ranked fruity candy?

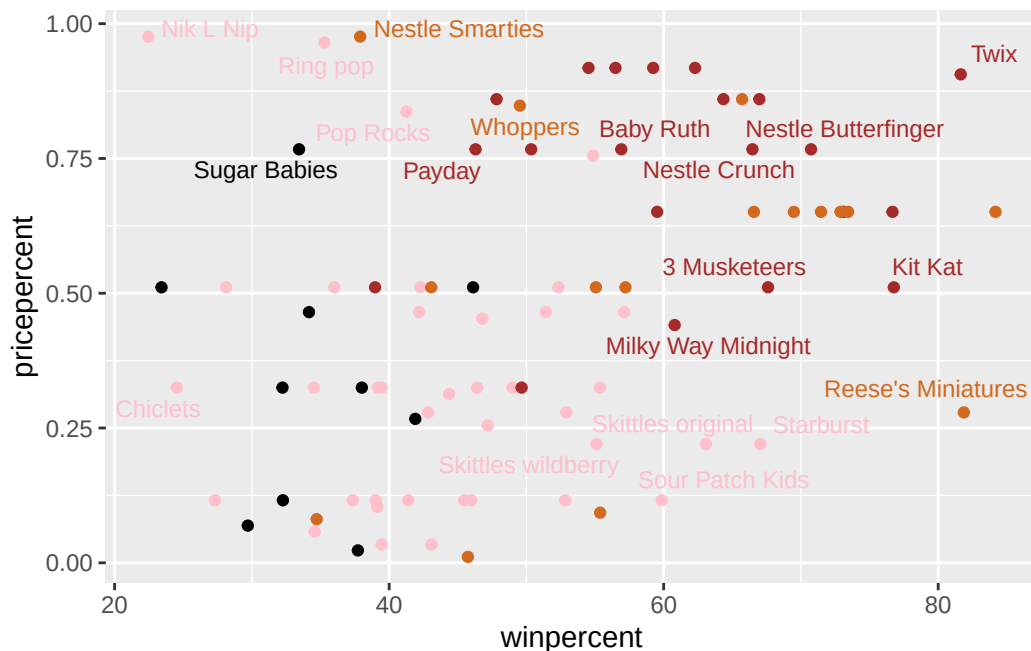
- The best ranked fruity candy is **Starburst**.

## Taking a look at Price point

The **pricepercent** variable records the percentile rank of the candy's price against all the other candies in the dataset. **Lower values are less expensive and higher values are more expensive**. In our ggplot, we will use the `geom_text_repel()` function from the **ggrepel** package to avoid overlapping labels which are hard to read.

```
library(ggrepel)
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of `winpercent` for the least money - i.e. offers the most bang for your buck?

- Reese's miniatures is the candy type is the highest ranked in terms of `winpercent` for the least money - i.e. offers the most bang for your buck.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

|                          | pricepercent | winpercent |
|--------------------------|--------------|------------|
| Nik L Nip                | 0.976        | 22.44534   |
| Nestle Smarties          | 0.976        | 37.88719   |
| Ring pop                 | 0.965        | 35.29076   |
| Hershey's Krackel        | 0.918        | 62.28448   |
| Hershey's Milk Chocolate | 0.918        | 56.49050   |

- The 5 most expensive candy types are given above and the least popular among these 5 is the Nik L Nip candy.

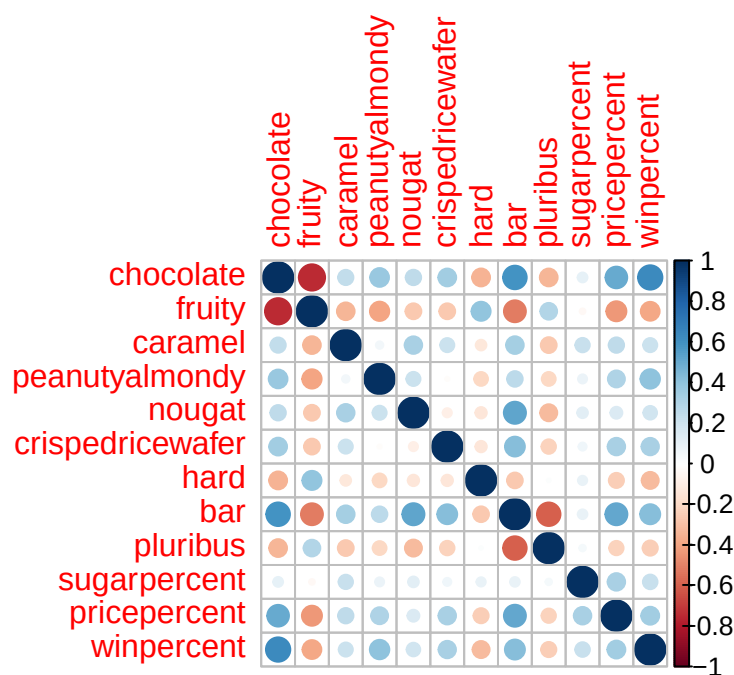
## 6. Exploring the correlation structure

We'll use correlation and view the results with the `corrplot` package to plot a correlation matrix.

```
library(corrplot)
```

`corrplot` 0.95 loaded

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

- From this plot, we see that the two variables that are anti-correlated (i.e have minus values) are **chocolate** and **fruity**.

Q23. Similarly, what two variables are most positively correlated?

- The two variables that are the most positively correlated are **chocolate** and **winpercent**.

## 7. Principal Component Analysis

Let's apply PCA using the `prcomp()` function to our candy dataset remembering to set the `scale=TRUE` argument.

```
pca <- prcomp(candy, scale= TRUE)
summary(pca)
```

Importance of components:

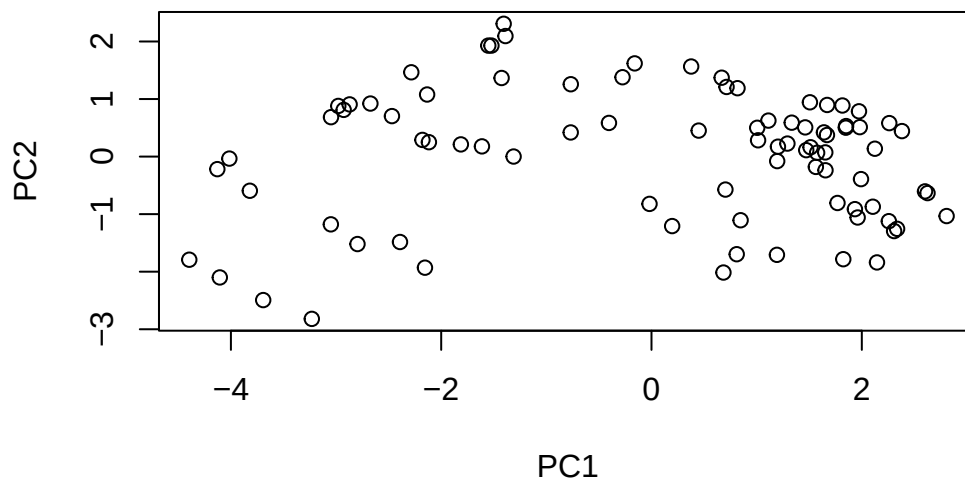
|                        | PC1    | PC2    | PC3    | PC4     | PC5    | PC6     | PC7     |
|------------------------|--------|--------|--------|---------|--------|---------|---------|
| Standard deviation     | 2.0788 | 1.1378 | 1.1092 | 1.07533 | 0.9518 | 0.81923 | 0.81530 |
| Proportion of Variance | 0.3601 | 0.1079 | 0.1025 | 0.09636 | 0.0755 | 0.05593 | 0.05539 |
| Cumulative Proportion  | 0.3601 | 0.4680 | 0.5705 | 0.66688 | 0.7424 | 0.79830 | 0.85369 |

|                        | PC8     | PC9     | PC10    | PC11    | PC12    |
|------------------------|---------|---------|---------|---------|---------|
| Standard deviation     | 0.74530 | 0.67824 | 0.62349 | 0.43974 | 0.39760 |
| Proportion of Variance | 0.04629 | 0.03833 | 0.03239 | 0.01611 | 0.01317 |
| Cumulative Proportion  | 0.89998 | 0.93832 | 0.97071 | 0.98683 | 1.00000 |

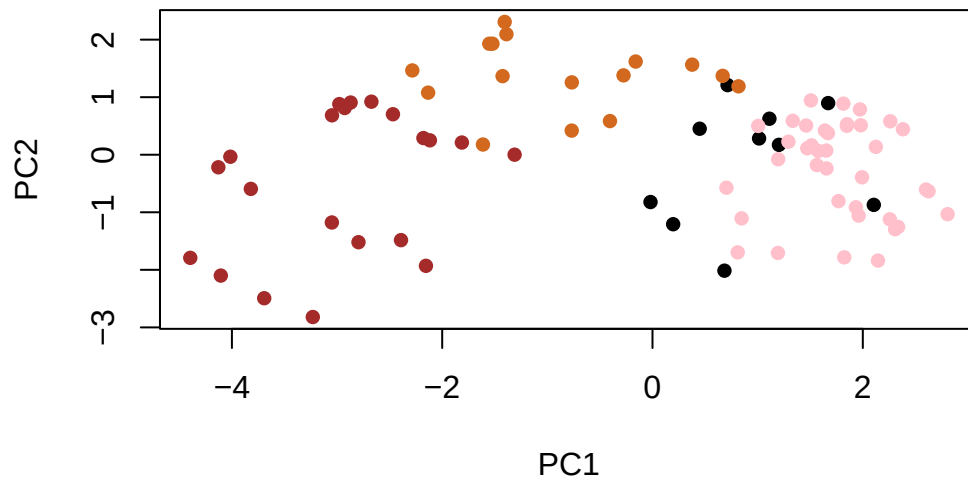
Now, we plot our main PCA score plot of PC1 vs PC2:

```
plot(pca$x[,1:2])
```



Changing the plotting color and adding some color:

```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



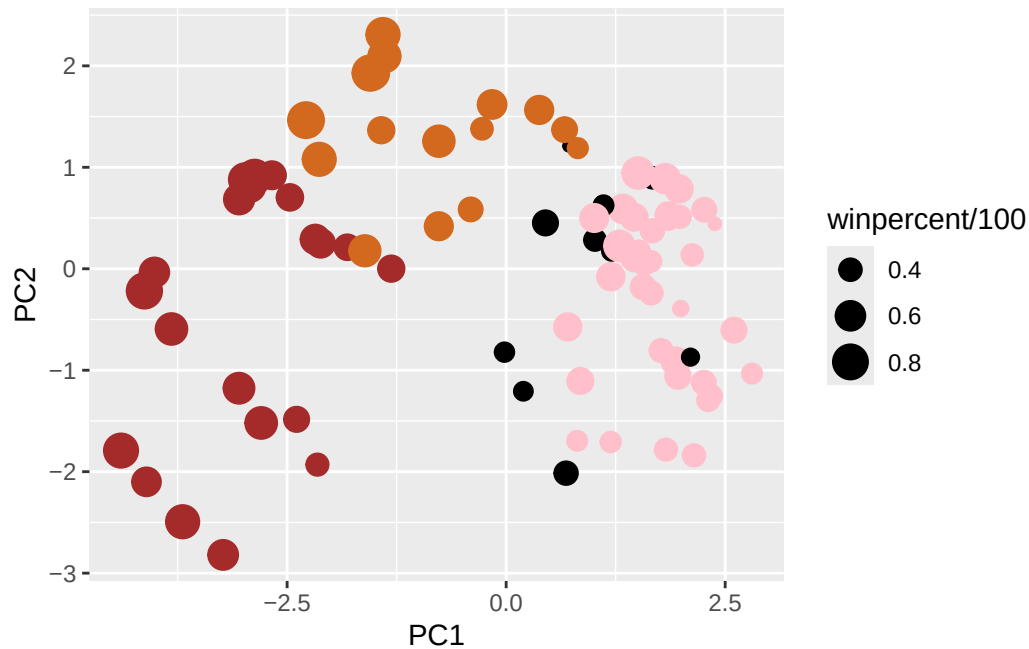
Making a new data-frame:

```
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +  
  aes(x=PC1, y=PC2,  
      size=winpercent/100,  
      text=rownames(my_data),  
      label=rownames(my_data)) +  
  geom_point(col=my_cols)
```

p





Again we can use the `ggrepel` package and the function `ggrepel::geom_text_repel()` to label up the plot with non overlapping candy names like. We will also add a title and subtitle like so:

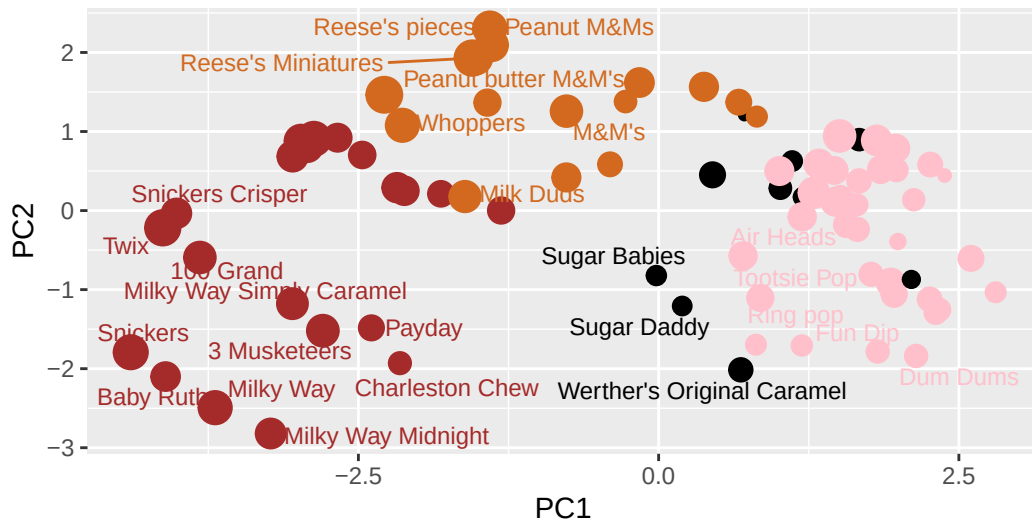
```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Warning: `ggrepel`: 59 unlabeled data points (too many overlaps). Consider increasing `max.overlaps`

## Halloween Candy PCA Space

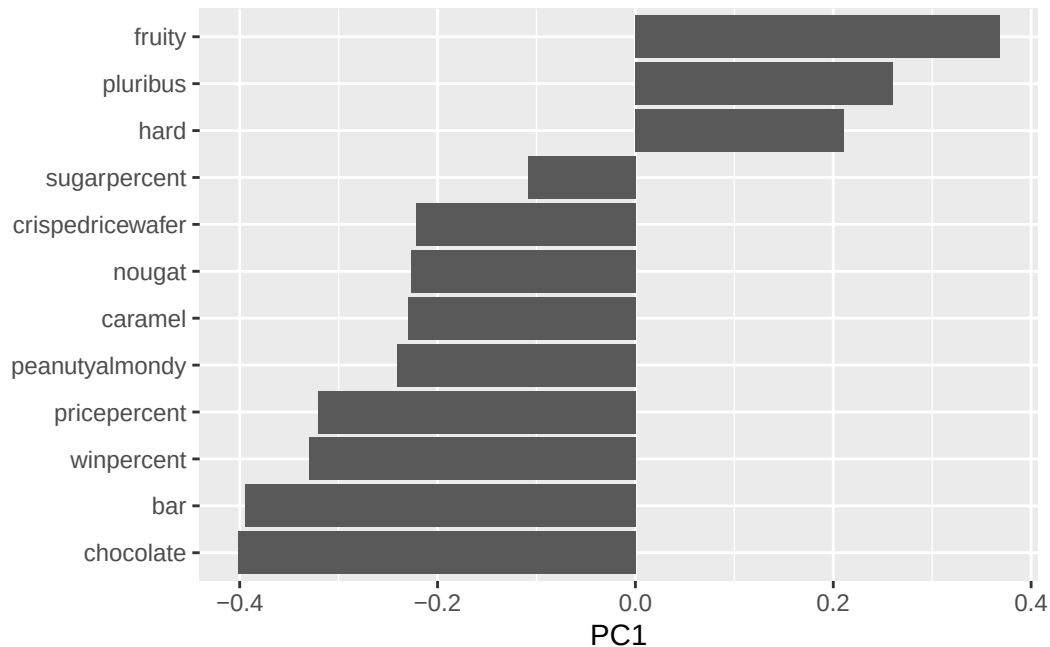
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
# install.packages("plotly")  
# library(plotly)  
# ggplotly(p)
```

```
ggplot(pca$rotation) +  
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +  
  geom_col() +  
  theme(axis.title.y = (element_blank()))
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

- The code is completed above to generate the plot above. The original variables that are picked up strongly by PC1 in the positive direction are **fruity**, **pluribus** and **hard**. Yes, this makes sense because it separates the two main different types of candies in the data set-candies with high PC1 scores which are fruity, pluribus and have a hard texture to the candies with negative PC1 scores that are chocolate and bar type. So, this plot captures the major split between the candy types. This relationship was highlighted previously in the correlation plot.

## 8. Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

- Based on our exploratory analysis, correlation findings, and PCA results, the combination of characteristics that appears to make a “winning” candy is **chocolate bar type often with things like peanut/almond/nougat/caramel**. These different analyses (visualization, correlation, PCA) support/complement each other in reaching

this conclusion as the plots including the ggplots produced consistently show the highest popularity (winpercent) for chocolate bar type of candy and low popularity of the fruity/pluribus/hard candy. The correlation plot also shows this result of the chocolate vs fruity being opposite and most vs least popular. Finally, the PCA brings all of the analysis together and gives us that clear split between the two main candy types- the chocolate bar type vs the fruity candy.